

Untitled

機械学習

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1 SetUp

- $\mu_{11} = K_{11} \times (1 - e)^2$
- $\mu_{01} = K_{11} \times e \times (1 - e)$
- $\mu_{01} = K_{11} \times e \times (1 - e) + K_{01} \times (1 - e)$

2 Moment

```
library(tidyverse)

data <- read_csv("Data/test.csv") |>
  mutate(
    A1 = if_else(Q1 == -50, 1, 0),
    A2 = if_else(Q2 == 100, 1, 0)
  )

Mu_11 <- if_else(data$A1 == 1 & data$A2 == 1, 1, 0) |> mean()
Mu_10 <- if_else(data$A1 == 1 & data$A2 == 0, 1, 0) |> mean()
Mu_01 <- if_else(data$A1 == 0 & data$A2 == 1, 1, 0) |> mean()
Mu_00 <- if_else(data$A1 == 0 & data$A2 == 0, 1, 0) |> mean()

E <- Mu_01 / (Mu_11 + Mu_01)

K_11 <- Mu_11 / ((1 - E)^2)

K_01 <- (Mu_10 - K_11 * (E) * (1 - E)) / (1 - E)

K_00 <- 0
```

3 Bayes

```
library(tidyverse)

data <- read_csv("Data/test.csv") |>
  mutate(
    A1 = if_else(Q1 == -50, 1, 0),
    A2 = if_else(Q2 == 100, 1, 0)
  )

parameter <- tibble(
  E = seq(0.01, 0.20, 0.01),
  K_11 = seq(0.80, 0.99, 0.01),
  K_10 = seq(0.01, 0.20, 0.01)
) |>
  expand(E, K_11, K_10)

N_11 <- if_else(data$A1 == 1 & data$A2 == 1, 1, 0) |> sum()
N_10 <- if_else(data$A1 == 1 & data$A2 == 0, 1, 0) |> sum()
N_01 <- if_else(data$A1 == 0 & data$A2 == 1, 1, 0) |> sum()
N_00 <- if_else(data$A1 == 0 & data$A2 == 0, 1, 0) |> sum()

parameter$L <- rep(NA |> as.numeric(), nrow(parameter))

for (i in 1:nrow(parameter)){
  parameter$L[i] <-
    ((parameter$K_11[i]*((1 - parameter$E[i])^2))^N_11)*
    ((parameter$K_11[i]*(1 - parameter$E[i])*parameter$E[i])^N_01)*
    ((parameter$K_11[i]*(1 - parameter$E[i])*parameter$E[i] +
parameter$K_10[i]*(1 - parameter$E[i]))^N_10)
}

parameter[parameter$L == max(parameter$L),]
```

```
# A tibble: 1 × 4
      E   K_11   K_10         L
<dbl> <dbl> <dbl>     <dbl>
1  0.05  0.99   0.2 0.000000468
```

4 ML

```
library(tidyverse)
```

```

data <- read_csv("Data/test.csv") |>
  mutate(
    A1 = if_else(Q1 == -50, 1, 0),
    A2 = if_else(Q2 == 100, 1, 0)
  )

parameter <- tibble(
  E = seq(0.01, 0.20, 0.01),
  K_11 = seq(0.80, 0.99, 0.01),
  K_10 = seq(0.01, 0.20, 0.01)
) |>
  expand(E, K_11, K_10)

N_11 <- if_else(data$A1 == 1 & data$A2 == 1, 1, 0) |> sum()

N_10 <- if_else(data$A1 == 1 & data$A2 == 0, 1, 0) |> sum()

N_01 <- if_else(data$A1 == 0 & data$A2 == 1, 1, 0) |> sum()

N_00 <- if_else(data$A1 == 0 & data$A2 == 0, 1, 0) |> sum()

parameter$L <- rep(NA |> as.numeric(), nrow(parameter))

for (i in 1:nrow(parameter)){
  parameter$L[i] <-
    (log(parameter$K_11[i]*((1 - parameter$E[i])^2))*N_11) +
    (log(parameter$K_11[i]*(1 - parameter$E[i])*parameter$E[i])*N_01) +
    (log(parameter$K_11[i]*(1 - parameter$E[i])*parameter$E[i] +
parameter$K_10[i]*(1 - parameter$E[i]))*N_10)
}

parameter[parameter$L == max(parameter$L),]

```

```

# A tibble: 1 × 4
      E  K_11  K_10      L
<dbl> <dbl> <dbl> <dbl>
1  0.05  0.99  0.2 -14.6

```